Problem 1. Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} x^2(2x + \frac{3}{2}) & 0 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

If Y=2X+3, find Var(Y).

Problem 2. Let X be a random variable with PDF $f_X(x)$. Find the PDF of the random variable $Y = |X|$

$$f_X(x) = \begin{cases} \frac{1}{3} & -2 < x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

Problem 3. Let $Z \sim N(0, 1)$. Find $P(0.5 < |Z - 12| < 1.5)$.

Problem 4. In this problem you may neglect the probability of twins.

(a) A family has 5 children. Find the probability that they have 2 boys and 3 girls.

(b) A family decides to have children until they have 3 girls. Find the probability that they have 2 boys.

(c) A family decides to have children until they have 3 girls. Let C be the total number of children in the family. Compute E[C] and V(C).

Problem 5. 60 scientists from 30 universities attend a conference. The conference includes a lunch in a cafeteria which has 30 tables each suitable for 2 people. The people seated for lunch at random. Let $X_j = 1$ if the scientists from university j seat together and $X_j = 0$ otherwise.

(a) Compute $E(X_1)$ and $V(X_1)$.

(b) Compute $Cov(X_1, X_2)$.

(c) Let $X = X_1 + X_2 + \cdots + X_{30}$ be the number of tables occupied by the people from the same university. Compute E[X] and V(X).

Problem 6. Let X be the number of Heads in 10 fair coin tosses.

(a) Find the conditional PMF of X, given that the first two tosses both land Heads.

(b) Find the conditional PMF of X, given that at least two tosses land Heads.

Problem 7. A book has n typos. Two proofreaders, Sarah and Hannah, independently read the book. Sarah catches each typo with probability p_1 and misses it with probability $1-p_1$, independently, and likewise for Hannah, who has probabilities p_2 of catching and $1-p_2$ of missing each typo. Let X_1 be the number of typos caught by Sarah, X_2 be the number caught by Hannah, and X be the number caught by at least one of the two proofreaders.

(a) Find the distribution of X.

(b) For this part only, assume that $p_1 = p_2$. Find the conditional distribution of X_1 given that $X_1 + X_2 = t$

Problem 8. Suppose that the time T that it takes to complete a Homework in hours follows a distribution with density

$$f(t) = \frac{2(t^2 + t)}{27}, \qquad 0 \leq t \leq 3$$

What is the probability that a randomly chosen student finishes the exam during the second hour of the exam. That is, calculate $P(1 < T < 2)$.

Problem 9. A factory produces high-quality glassware. When the furnace temperature is correctly maintained, the glass defect rate is 5%. However, there is a 10% chance that the temperature control system malfunctions, in which case the defect rate rises to 30%.

(a) Suppose a randomly selected glass item from the production line is found to be defective. What is the probability that the furnace temperature control had malfunctioned?

(b) A random sample of 20 glass items was inspected, and 4 of them were found to be defective. What is the probability that the furnace temperature control had malfunctioned?

Problem 10. Each of $n \geq 2$ people puts his or her name on a slip of paper (no two have the same name). The slips of paper are shuffled in a hat, and then each person draws one (uniformly at random at each stage, without replacement). Find the mean and standard deviation of the number of people who draw their own names.